



Veridian: A Technical, Legal, and Economic Architecture for Tokenized Solar Infrastructure in South Africa

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ABSTRACT

Tokenizing Real-World Assets (RWAs) fundamentally changes how financial markets operate. According to a report by the Boston Consulting Group [1], the tokenization of illiquid assets is expected to reach \$16 trillion by 2030. This shift brings physical assets onto the blockchain, making them secure, easy to trade, and unlocking new investment opportunities that were previously completely out of reach. This report provides an interdisciplinary analysis of a purpose-built architecture that lets people buy and trade small shares of real-world assets. By integrating technical infrastructure, smart contract engineering, dynamic economic models, and multi-jurisdictional legal frameworks, we contextualize this architecture through distributed solar generation in South Africa. A country where households and small businesses face prohibitive upfront installation costs while outside investors lack transparent mechanisms to track asset performance. To bridge this gap, we present Veridian, a proof-of-concept platform for financing solar equipment under an Equipment-as-a-Service model. Investment capital is pooled in USDC and held in a 3-of-5 multisignature escrow until installation and legal-ownership conditions are confirmed. Fractional interests are represented by ERC-3643 tokens that restrict transfers to verified investors. Collected revenue is distributed through a fixed waterfall: 75% to investors, 8% to operations and maintenance, 7% to insurance and reserves, 5% to expansion, and 5% to platform operations. The prototype demonstrates the technical workflow but simulates the physical TPM and retains important legal, operational, and payment dependencies.

1 INTRODUCTION

1.1 Bridging South Africa's Solar-Financing Gap

In South Africa, two bodies, Eskom and the National Energy Regulator of South Africa (NERSA), are primarily responsible for the supply and regulation of electricity. NERSA is meant to balance Eskom's financial survival against consumer protection [6]. However, in practice, that balance is never struck. Rather than regulating Eskom effectively, the regulator is occupied with repeated litigation and spends much of its resources defending decisions in court. For instance, in 2020, the High Court overturned NERSA's pricing model after finding that it had unlawfully stripped a R69 billion government bailout from Eskom's budget, undoing the state's efforts to rescue the utility from a massive cash crisis. Consequently, regulatory gridlock persists, leaving consumers facing surging costs.

Regulated Eskom tariffs have risen well above inflation for more than a decade (Figure 1). Average tariffs increased by approximately 12.7% in 2025 against consumer-price inflation of about 3.8%, and further above-inflation increases of 8.76% (2026) and 9.19% (2027) have already been approved [8, 4]. Rooftop solar can reduce dependence on the centralized grid, but it remains capital-intensive: a household that could afford a monthly energy-service payment may still be unable or unwilling to finance the full purchase and installation cost upfront. This produces a financing mismatch. Solar operators need capital before equipment can be acquired and installed, whereas households receive economic benefit gradually. Investors could supply this capital, but they need evidence that the assets exist, generate electricity, remain operational, and produce collectible revenue. Representing an asset on-chain does not by itself prove its physical condition or guarantee that revenue has been collected.

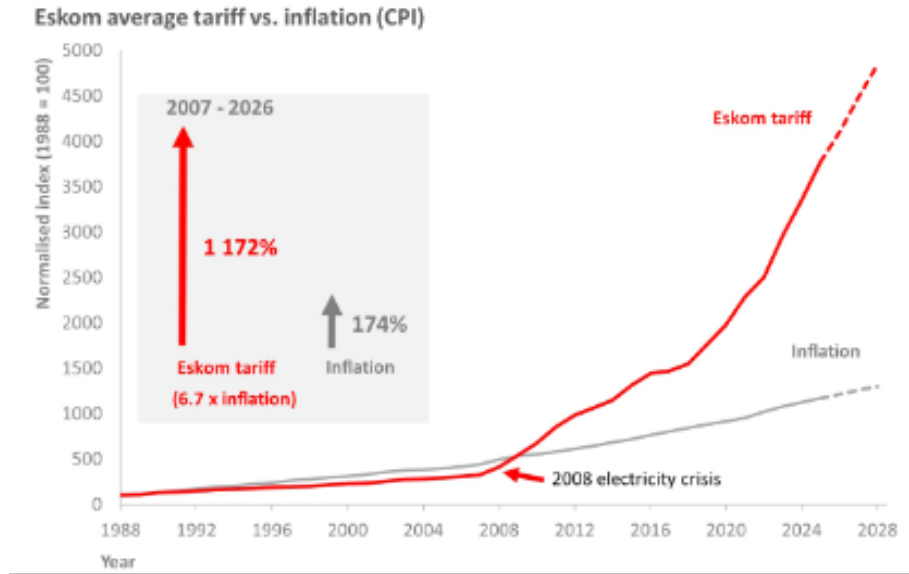


Fig. 1. Eskom average standard-tariff increases vs. CPI inflation, 1988–2026, with approved increases to 2028. Source: PowerOptimal (2026).

1.2 Problem Statement and Research Question

The project addresses three connected problems:

1. Investment capital must remain protected until a genuine installation and the corresponding legal rights exist.
2. Fractional ownership must comply with identity and transfer restrictions.
3. Physical generation data must be connected to financial settlement without allowing a platform administrator to alter readings or payouts unilaterally.

These motivate our research question: *How can a blockchain-based prototype connect verified solar generation to compliant fractional financing and automated investor settlement while making its remaining off-chain trust assumptions explicit?*

1.3 Contributions and Scope

We make four contributions: (1) an end-to-end financing and settlement lifecycle for South African solar assets; (2) an ERC-3643 token and multisignature escrow architecture; (3) an ECDSA-authenticated telemetry and Chainlink settlement pipeline; and (4) a working prototype and evaluation covering identity controls, escrow release, telemetry validation, and revenue distribution. This project is a technical proof of concept. It does not represent a public securities offering, a licensed investment product, a legal opinion, or a completed solar deployment.

2 CONTEXT AND DESIGN REQUIREMENTS

2.1 Stakeholders and Operating Model

Veridian coordinates four stakeholder groups and three technical functions. The household or lessee receives the solar installation and pays an Equipment-as-a-Service fee. The solar operator procures, installs, maintains, and inspects the equipment. An asset-holding special-purpose vehicle (SPV) owns the equipment and defines the contractual rights of investors. Verified investors provide the acquisition capital and receive ERC-3643 tokens, while approved identity issuers assess their eligibility and issue the required identity claims.

The technical workflow supports these stakeholders. The Veridian backend verifies telemetry and calculates billable revenue; Chainlink delivers authorized reports to the blockchain, and smart contracts enforce escrow conditions, token-transfer restrictions, and revenue distributions. Investor cash flow originates from the household's collected service payment, not from electricity generation itself. Telemetry verifies performance and supports billing; it does not create revenue directly.

2.2 Design Requirements

We derived six requirements that the architecture must satisfy, summarized in Table 1.

ID	Requirement	Description
R1	Capital protection	Release funds only after required signers confirm installation and contractual conditions.
R2	Compliant ownership	Token minting and transfers succeed only with valid ERC-3643 identity claims.
R3	Telemetry integrity	Accept only correctly formatted, signed, and non-replayed telemetry readings.
R4	Deterministic yield	Input data produces exact 75%, 8%, 7%, 5%, and 5% fee splits deterministically.
R5	Restricted payout	Only designated Chainlink forwarder submits distribution reports; no double payouts.
R6	Auditability	Material contract actions emit on-chain events; telemetry retains timestamped records.

Table 1. Design requirements for the Veridian prototype.

2.3 Trust and Threat Model

Veridian reduces reliance on a single platform administrator but does not eliminate trust. The design assumes that identity issuers perform eligibility checks correctly, at least 3 of the 5 escrow signers act honestly, the solar operator installs equipment properly, the SPV's legal records are enforceable, and the payment provider accurately confirms collected household revenue.

Principal threats include unverified investors attempting to acquire tokens, insufficient signers releasing escrow funds, operators modifying telemetry, replayed valid readings, unauthorized payout triggers, or double distribution of a billing cycle. ECDSA protects payload integrity after signing, but it cannot prove physical meter accuracy or household bill payment. These limitations explain why the architecture retains physical inspections, payment confirmation, insurance, and financial reserves.

3 PROPOSED VERIDIAN ARCHITECTURE

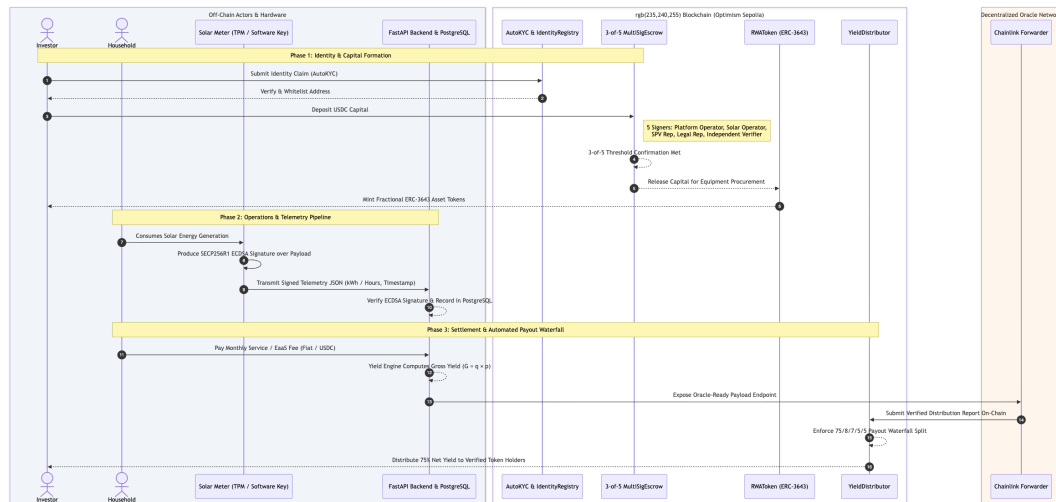


Fig. 2. Veridian System Architecture and Lifecycle Sequence Diagram [16].

3.1 Asset Formation and Escrow

Each installation pool is associated with an asset-holding SPV. Investors deposit USDC into a 3-of-5 multisignature escrow (`MultiSigEscrow.sol`). Five independent roles hold signing keys: platform operator, solar operator, SPV representative, legal/compliance representative, and independent verifier. At least three must approve a release. After confirmation, capital is released for installation, and investors receive tokenized interests linked to the SPV.

3.2 Identity and Transfer Controls

Veridian implements a compliance-focused token design inspired by ERC-3643 principles [3]. The design separates identity and compliance logic: an Identity Registry records address verification, trusted issuers define authorized claim issuers, and the token checks eligibility before any transfer. Minting to unverified addresses is rejected, addresses can be frozen, and compliance agents retain forced-transfer capabilities. Identity issuance is currently mocked via an AutoKYC flow.

3.3 Telemetry and Oracle Pipeline

Meters produce telemetry containing asset IDs, timestamps, operating hours, utilization rates, and power consumption. The backend verifies an ECDSA signature over a canonical JSON payload using software keys (simulating physical TPM hardware). Verified readings are stored in PostgreSQL. The yield engine aggregates readings per billing cycle, combined with contracted energy tariffs.

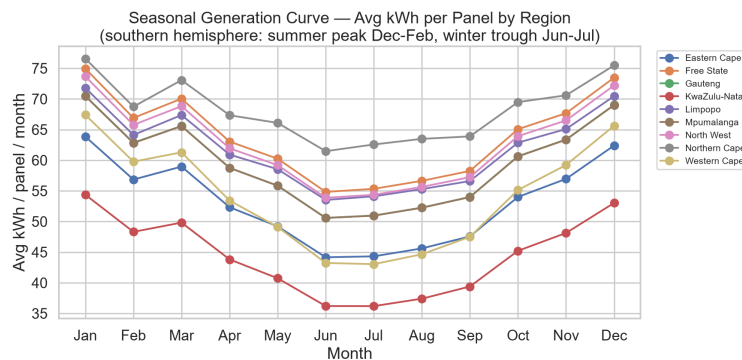


Fig. 3. Average monthly kWh generation per panel across South African provinces, illustrating the seasonal variance captured by Veridian telemetry.

Figure 3 illustrates the seasonal baseline curves that the telemetry pipeline processes. Since generation peaks in the southern hemisphere summer (Dec–Feb) and troughs in winter (Jun–Jul), the yield engine must dynamically calculate distributions based on actual verified operating hours rather than flat monthly estimates.

3.4 Revenue Calculation and Distribution

Gross yield is defined as $G = q \times p$, where q is verified usage or operating hours and p is the unit price. Net yield distributed to investors equals $0.75G$, with the remaining 25% allocated to maintenance (8%), reserves/insurance (7%), expansion (5%), and platform fees (5%).

4 PROTOTYPE IMPLEMENTATION

4.1 Smart Contracts

Built using Foundry and OpenZeppelin libraries. `MultiSigEscrow.sol` enforces 3-of-5 threshold release. `RWAFactory.sol` deploys `RWAToken.sol` (extending OpenZeppelin ERC20) alongside governance and liquidity pool contracts. `RWAToken` overrides transfer hooks to enforce `IdentityRegistry.isVerified()` and freeze checks. `YieldDistributor.sol` restricts distribution reports to an authorized forwarder.

4.2 Backend and Oracle Integration

The backend is an asynchronous FastAPI service backed by PostgreSQL. Signature verification uses Python’s cryptography library (SECP256R1/SHA-256) over canonical JSON payloads. The yield engine computes deterministic fee splits. Chainlink integration is exposed via an API-key-authenticated JSON endpoint ready for Decentralized Oracle Network (DON) delivery.

4.3 Frontend and User Flow

Three routed portals are implemented in Next.js/TypeScript using `wagmi` and `viem`: Household/Lessee, Operator/Admin, and Investor. Direct contract reads/writes go on-chain, while the FastAPI backend provides telemetry, asset, and billing analytics.

4.4 Implementation Status

Table 2 summarizes the implementation status of all core architecture components.

Component	Status	Evidence & Remaining Work
ERC-3643 Flow	Partial	Tests pass for verified hooks; needs claims-based registry integration.
3-of-5 Escrow	Implemented	Tests pass for threshold release; lacks on-chain link to physical installation.
75/8/7/5/5 Split	Partial	Forwarder-gated payout works; needs full 4-category split on-chain.
Opt. Sepolia	Proposed	Deployment script exists; requires live broadcast.
TPM Telemetry	Simulated	Software ECDSA verification active; physical TPM integration required.
Yield Engine	Partial	Deterministic fee calculated on operating hours; requires metered kWh billing.
Chainlink Report	Simulated	Static JSON endpoint active; live DON forwarder transaction needed.
Frontend Portals	Partial	Core routes implemented via wagmi/viem; full integration pending.
AutoKYC Issuer	Mocked	Simplified issuer bypasses manual checks; licensed integration required.
Midnight Layer	Proposed	Selective disclosure layer proposed for future privacy.
Fiat/USDC Rail	Proposed	Requires real payment rails and fiat conversion.
SA SPV Legal	Proposed	Requires formal incorporation and professional legal review.

Table 2. Implementation status by component.

5 EVALUATION

5.1 Method

Our evaluation checks whether the prototype enforces identity restrictions, multisig thresholds, telemetry tampering detection, and deterministic yield distribution under controlled execution.

5.2 Functional Test Results

We executed 10 functional tests (T1–T10). The prototype successfully passed all checks, including verified-only minting and transfer hooks (T1–T3), 3-of-5 threshold escrow release (T4–T5), ECDSA signature verification and payload tampering detection (T6–T7), and restricted payouts with correct 75/8/7/5/5 fee distribution (T8–T10).

5.3 Interpretation

Passing tests demonstrate rule enforcement under controlled conditions. They do not prove physical meter security or legal claim enforceability. Push-based distributions via dynamic arrays present gas scalability limits (local testing recorded 24,754 to 85,449 gas depending on holder count); a pull-based or claim-accumulator pattern will be required for production.

6 FEASIBILITY ANALYSIS

6.1 Worked Solar Case

Using a baseline residential solar asset cost $C = \$5,000$, monthly generation q [5], price p [4], performance degradation d [2], downtime δ , and collection rate c , gross annual revenue is modeled as:

$$G_{\text{annual}} \approx 12 \times q \times p \times (1 - \delta) \times c \times (1 - d) \quad (1)$$

Net investor payout equals $0.75 \times G_{\text{annual}}$. Investor returns depend directly on collected household cash flow, not raw physical generation.

As illustrated in Figure 4, the net distributable yield is highly sensitive to the geographic deployment of the asset. Regions with superior solar irradiance, such as the Northern Cape, achieve significantly higher yields compared to coastal regions like KwaZulu-Natal. Furthermore, because the upfront capital cost (C) is fixed at the time of installation, the compounding effect of Eskom’s above-inflation tariff increases (driving p higher) causes the investor yield to scale aggressively over the asset’s lifecycle, projecting steady-state net yields exceeding 20% by the late 2020s.

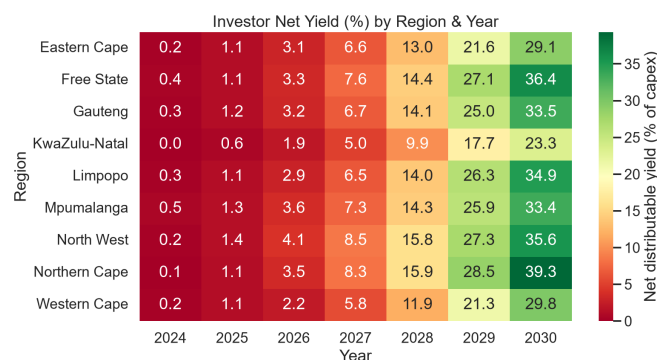


Fig. 4. Projected Investor Net Yield (%) by South African Province (2024–2030). Yields compound over time as regulated Eskom tariffs outpace the fixed initial capital expenditure.

6.2 Legal and Regulatory Position

Key South African legal frameworks governing this architecture include SPV incorporation under the Companies Act [11], financial product classification under the Financial Sector Regulation Act [14], FICA KYC/AML compliance [9], Consumer Protection and National Credit Acts [10, 12], POPIA data handling [13], and SARS crypto-asset taxation guidance [15].

6.3 Risk Allocation

The 75/8/7/5/5 waterfall balances investor capital recovery (75%) against routine O&M (8%), component reserve/insurance (7%), pool expansion (5%), and platform software/oracle maintenance (5%).

The long-term viability of the 75/8/7/5/5 waterfall relies on achieving economies of scale. Figure 5 models a 6-year growth trajectory for the Veridian platform. As cumulative panels financed scale past 1,000 units, the total annual generation approaches 1 Megawatt-hour (MWh), driving robust platform revenue that safely covers operations, maintenance, and insurance reserves. Crucially, the model demonstrates a parallel growth in household bill savings, ensuring that the Equipment-as-a-Service model remains economically attractive to the lessee, which in turn minimizes the risk of payment defaults.

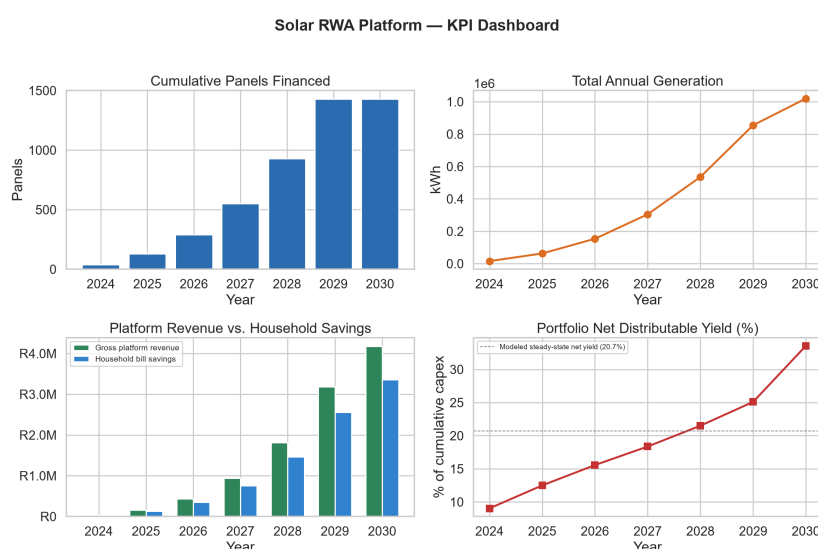


Fig. 5. Veridian Platform 6-Year Projections: Cumulative scaling, generation output, and economic split between platform revenue and household savings.

7 LIMITATIONS AND FUTURE WORK

Primary prototype limitations include simulated TPM keys, lack of live DON testnet transactions, custom whitelist logic rather than full T-REX/ERC-3643 implementation, dev-mode backend bypasses, and un-audited smart contracts. Prioritized future work involves completing full ERC-3643 claims integration, hardening contract access controls, executing live Chainlink testnet calls, integrating hardware TPMs, and conducting a pilot deployment with an incorporated South African SPV.

8 DISCUSSION AND CONCLUSION

Veridian demonstrates how blockchain can structure and coordinate the financing lifecycle of solar projects. Key elements include secure capital formation, restricted token ownership, auditable telemetry processing, and automated revenue distribution. Rather than claiming to eliminate all intermediaries, Veridian's primary focus is to clarify which decisions can be automated via software and which still rely on physical operators, legal entities, payment providers, and regulators. In short, the project serves as a practical proof of concept for system integration. Ultimately, Veridian provides a solid foundation to explore how fractional investment can lower financial barriers for solar infrastructure in South Africa. Beyond solar panels, this model offers a blueprint for general global climate finance and shows that blockchain's true value lies not in replacing human actors, but in creating seamless, verifiable channels for international capital.

AUTHOR CONTRIBUTIONS

All authors conceived the project idea and business model. J.G. Scherler wrote the abstract and led the introduction, discussion, and conclusion. S.D. Wu led context, requirements, and non-technical limitations. A. Idirisova led architectural/implementation documentation for backend and smart contracts. H. Chaabouni led frontend documentation and legal/economic feasibility. K. Sasinthiran developed the complete codebase (smart contracts, backend, frontend) and led technical evaluation. All authors reviewed and approved the final document.

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